

Agentic AI

Assignment 3

Author: Prashant Meena (24b0949)

July 1, 2026



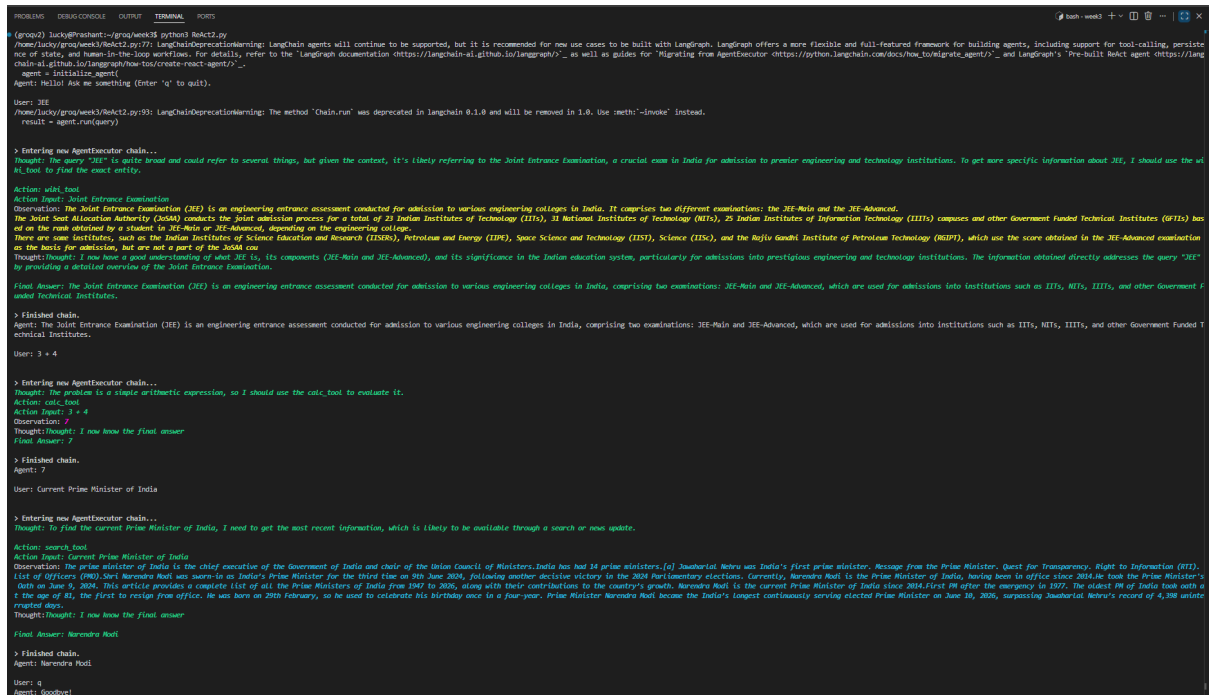
Figure 1: Chatbot Image

Solution 1

Link of my Google Drive (All 3 Questions)

Code File

I have implemented a ReAct agent which has access to 3 tools. It uses Wikipedia for info related to facts or persons or places and if we need to search something that might not be on wikipedia then it uses DUCKDUCKGO Search. It can also use Calculator btw.



```
PROBLEMS  DEBUG CONSOLE  OUTPUT  TERMINAL  PORTS
(lmcp2) lucky@vachant:~/groq/week18$ python3 ReAct2.py
/home/lucky/groq/week18/ReAct2.py:77: LangChainDeprecationWarning: LangChain agents will continue to be supported, but it is recommended for new use cases to be built with LangGraph. LangGraph offers a more flexible and full-featured framework for building agents, including support for tool-calling, persistence of state, and human-in-the-loop workflows. For details, refer to the 'LangGraph documentation https://langchain-ai.github.io/langgraph/' as well as guides for 'Migrating from AgentExecutor https://python.langchain.com/docs/how_to/migrate_agent/' and LangGraph's 'Pre-built ReAct agent https://langchain-ai.github.io/langgraph/how-to/create-react-agent/'...
  agent = initialize_agent(
Agent: Hello! Ask me something (Enter 'q' to quit).

User: JEE
/home/lucky/groq/week18/ReAct2.py:93: LangChainDeprecationWarning: The method 'Chain.run' was deprecated in LangChain 0.1.0 and will be removed in 1.0. Use 'meth.__invoke__' instead.
  result = agent.run(query)

> Entering new AgentExecutor chain...
Thought: The query "JEE" is quite broad and could refer to several things, but given the context, it's likely referring to the Joint Entrance Examination, a crucial exam in India for admission to premier engineering and technology institutions. To get more specific information about JEE, I should use the web tool to find the most entity.
Action: web_tool
Action Input: Joint Entrance Examination
Observation: The Joint Entrance Examination (JEE) is an engineering entrance assessment conducted for admission to various engineering colleges in India. It comprises two different examinations: the JEE-Main and the JEE-Advanced. The Joint Test Allocation Authority (JEEAA) conducts the Joint admission process for a total of 23 Indian Institutes of Technology (IITs), 31 National Institutes of Technology (NITs), 25 Indian Institutes of Information Technology (IIITs) campuses and other Government Funded Technical Institutes (GFTIs) based on the rank obtained by a student in JEE-Main or JEE-Advanced, depending on the engineering college. There are some institutes, such as the Indian Institutes of Science Education and Research (IISERs), Petroleum and Energy (IITPE), Space Science and Technology (IIIST), Science (IISc), and the Rajiv Gandhi Institute of Petroleum Technology (RGPII), which use the score obtained in the JEE-Advanced examination as the basis for admission, but are not a part of the JEEAA con
Thought: I now have a good understanding of what JEE is, its components (JEE-Main and JEE-Advanced), and its significance in the Indian education system, particularly for admissions into prestigious engineering and technology institutions. The information obtained directly addresses the query "JEE" by providing a detailed overview of the Joint Entrance Examination.
Final Answer: The Joint Entrance Examination (JEE) is an engineering entrance assessment conducted for admission to various engineering colleges in India, comprising two examinations: JEE-Main and JEE-Advanced, which are used for admissions into institutions such as IITs, NITs, IIITs, and other Government Funded Technical Institutes.

> Finished chain.
Agent: The Joint Entrance Examination (JEE) is an engineering entrance assessment conducted for admission to various engineering colleges in India, comprising two examinations: JEE-Main and JEE-Advanced, which are used for admissions into institutions such as IITs, NITs, IIITs, and other Government Funded Technical Institutes.

User: 3 + 4

> Entering new AgentExecutor chain...
Thought: The problem is a simple arithmetic expression, so I should use the calc tool to evaluate it.
Action: calc_tool
Action Input: 3 + 4
Observation: 7
Thought: I now know the final answer
Final Answer: 7

> Finished chain.
Agent: 7

User: Current Prime Minister of India

> Entering new AgentExecutor chain...
Thought: To find the current Prime Minister of India, I need to get the most recent information, which is likely to be available through a search or news update.
Action: search_tool
Action Input: Current Prime Minister of India
Observation: The prime minister of India is the chief executive of the Government of India and chair of the Union Council of Ministers. India has had 14 prime ministers.(a) Jawaharlal Nehru was India's first prime minister. Message from the Prime Minister. Quest for Transparency. Right to Information (RTI). List of Officers (PMO) Sir Narendra Modi has served as India's Prime Minister for the third time on 9th June 2024, following another decisive victory in the 2024 Parliamentary elections. Currently, Narendra Modi is the Prime Minister of India, having been in office since 2014.He took the Prime Minister's oath on June 5, 2024. This article provides a complete list of all the Prime Ministers of India from 1947 to 2026, along with their contributions to the country's growth. Narendra Modi is the current Prime Minister of India since 2014.First PM after the emergency in 1977. The oldest PM of India took oath at the age of 81, the first to resign from office. He was born on 29th February, so he used to celebrate his birthday once in a four-year. Prime Minister Narendra Modi became the India's longest continuously serving elected Prime Minister on June 10, 2026, surpassing Jawaharlal Nehru's record of 4,398 united ruled days.
Thought: I now know the final answer
Final Answer: Narendra Modi

> Finished chain.
Agent: Narendra Modi

User: q
Agent: Goodbye!
```

Figure 2: Terminal Output

```
1 from langchain_community.tools import DuckDuckGoSearchRun
2 from langchain.agents import initialize_agent, AgentType
3 from langchain_groq import ChatGroq
4 from langchain.tools import tool
5 from dotenv import load_dotenv
6 import requests
7 import os
8
9 load_dotenv()
10 GROQ_API_KEY = os.getenv("GROQ_API_KEY")
11
12 llm = ChatGroq(
13     model="llama-3.3-70b-versatile",
14     api_key=GROQ_API_KEY,
15 )
16
17 search = DuckDuckGoSearchRun()
18
19 @tool
20 def calc_tool(expr: str) -> str:
```

```

21     """Use only and only for arithmetic expressions like 2+2 or
22     25*(3+2)."""
23     return str(eval(expr))
24
25 @tool
26 def search_tool(query: str) -> str:
27     """Use for current events, weather, recent facts, news."""
28     return search.run(query)
29
30 @tool
31 def wiki_tool(query: str) -> str:
32     """
33     Use for exact entity names only.
34     Examples:
35     Napoleon
36     Adolf Hitler
37     World War II
38     """
39     try:
40         headers = {
41             "User-Agent": "ReAct/1.0"
42         }
43
44         search_url = "https://en.wikipedia.org/w/api.php"
45
46         params = {
47             "action": "query",
48             "prop": "extracts",
49             "exintro": True,
50             "explaintext": True,
51             "titles": query,
52             "format": "json"
53         }
54
55         r = requests.get(search_url, params=params, headers=
56             headers, timeout=10)
57         r.raise_for_status()
58
59         data = r.json()
60         pages = data["query"]["pages"]
61
62         page = next(iter(pages.values()))
63
64         if "missing" in page:
65             return f"No Wikipedia page found for: {query}"
66
67         text = page.get("extract", "").strip()
68
69         if not text:
70             return "No summary available."

```

```

70         return text[:1000]
71
72     except Exception as e:
73         return "Wikipedia API failed."
74
75 tools = [search_tool, wiki_tool, calc_tool]
76
77 agent = initialize_agent(
78     tools=tools,
79     llm=llm,
80     agent=AgentType.ZERO_SHOT_REACT_DESCRIPTION,
81     verbose=True,
82     handle_parsing_errors=True
83 )
84
85 print("Agent: Hello! Ask me something (Enter 'q' to quit).")
86 while True:
87     query = input("\nUser: ").strip()
88
89     if query == 'q':
90         print("Agent: Goodbye!")
91         break
92
93     result = agent.run(query)
94     print("Agent:", result)

```

Solution 2

I have implemented a LangGraph based Research agent which uses a local ollama based model (slower and weaker than groq based models).

```

1
2 from typing import TypedDict
3 from langgraph.graph import StateGraph, END
4 from langchain_ollama import ChatOllama
5
6 llm = ChatOllama(model="gemma2:2b")
7
8 class AgentState(TypedDict):
9     question: str
10    answer: str
11    score: int
12    iterations: int
13
14 def generate(state):
15     print("Entered generate")
16     prompt = f"Answer: {state['question']}"
17     response = llm.invoke(prompt)
18     return {
19         "answer": response.content,

```

```

20         "iterations": state["iterations"] + 1
21     }
22
23 def evaluate(state):
24     print("Entered evaluate")
25     answer = state["answer"]
26     words = len(answer.split())
27
28     score = state["score"]
29     if words > 100:
30         score += 1
31
32     return {"score": score}
33
34 def refine(state):
35     print("Entered refine")
36     response = llm.invoke(f"Improve this:\n{state['answer']}")
37     return {"answer": response.content}
38
39 def route(state):
40     if state["score"] >= 1:
41         print("Routed to end")
42         return "end"
43     print("Routed to refine")
44     return "refine"
45
46 builder = StateGraph(AgentState)
47
48 builder.add_node("generate", generate)
49 builder.add_node("evaluate", evaluate)
50 builder.add_node("refine", refine)
51
52 builder.set_entry_point("generate")
53
54 builder.add_edge("generate", "evaluate")
55 builder.add_edge("refine", "evaluate")
56
57 builder.add_conditional_edges(
58     "evaluate",
59     route,
60     {
61         "end": END,
62         "refine": "refine"
63     }
64 )
65
66 graph = builder.compile()
67
68 query = input("Enter your topic: ")
69 initial_state = {

```

```

70     "question": query,
71     "answer": "",
72     "score": 0,
73     "iterations": 0
74 }
75
76 result = graph.invoke(initial_state)
77 print(result)

```

```

(venv) prashantdp@Prashant:~/Desktop/Code/Projects/LangGraph/Agentic_AI$ python3 Workflow.py
Enter your topic: Who is Iron Man ?
Entered generate
Entered evaluate
Routed to end
{'question': 'Who is Iron Man ?', 'answer': 'Iron Man, real name **Tony Stark**, is a fictional superhero appearing in American comic books published by Marvel Comics. \n\nHere's a breakdown of who he is:\n\n**A billionaire industrialist:** Tony Stark was born into wealth and privilege, becoming an incredibly successful weapons manufacturer, but this led him to develop a sense of remorse for his role in the arms trade.\n\n**Inventor & Engineer:** He creates a special suit of armor that gives him superhuman abilities, transforming him from a playboy with a troubled past into a hero who fights crime and protects the world.\n\n**A complex character:** Iron Man grapples with his own conscience, often battling internal struggles while using his powers for good. He's known for his witty humor and sarcastic remarks. \n\n**His abilities include:**\n\n**Advanced technology:** The iconic suit features repulsor beams, flight capabilities, a variety of armor suits, and more.\n\n**Genius-level intellect:** Tony is known for his intelligence and problem-solving skills, crucial in designing and utilizing the Iron Man suit and its vast arsenal. \n\n\n**Iron Man's impact:**\n\n**Cultural icon:** He has become one of the most iconic superheroes in pop culture, inspiring numerous adaptations across comics, movies, television, video games, and more.\n\n**Symbol of innovation & morality:** He represents the power of technology for good and the importance of facing personal demons to become a better person.\n\n\nIf you're interested in learning more about Iron Man, there are countless resources available online, including official comic books, websites like Marvel.com, and other entertainment platforms where they have produced various adaptations. \n', 'score': 1, 'iterations': 1}
(venv) prashantdp@Prashant:~/Desktop/Code/Projects/LangGraph/Agentic_AI$ python3 Workflow.py
Enter your topic: How did pop culture spread in India ?
Entered generate
Entered evaluate
Routed to end
{'question': 'How did pop culture spread in India ?', 'answer': '## How Pop Culture Spreads in India: A Complex & Evolving Landscape \n\nPop culture\'s journey through India is a fascinating story of adaptation, innovation, and cultural shifts. Here's a look at some key factors driving its growth:\n\n**Traditional Influences:**\n\n**Local Media:** Regional languages and regional film industries played an important role in shaping pop culture. Bollywood, with its iconic stars and narratives, remains a dominant force.\n\n**Religious Festivals & Traditions:** Religious festivals like Diwali, Holi, Eid, etc., are vibrant occasions for sharing pop culture experiences through music, dance, storytelling, and more.\n\n**Folk Music & Dance:** Traditional forms of folk music and dance continue to influence popular music genres and contemporary performances.\n\n\n**Modern Influences:**\n\n**Globalization:** The rise of global media networks like MTV, Netflix, and YouTube has introduced international content, pop stars, trends, and ideas into Indian households. \n\n**Social Media:** Social platforms have become key avenues for disseminating pop culture - from celebrity gossip and memes to viral challenges and dance trends. \n\n**Mobile Phones & Streaming:** India boasts a vast population of mobile phone users and a thriving streaming landscape. This gives rise to personalized content consumption, impacting music tastes, movie preferences, and social media trends.\n\n\n**Specific Examples:**\n\n**Music:** Bollywood films and pop stars like AR Rahman, Priyanka Chopra Jonas, Shah Rukh Khan, and others have global influence on Indian music trends.\n\n**Television:** Reality shows, dramas, and sitcoms resonate with a large audience, often mirroring themes of family, love, and societal issues. \n\n**Social Media & Memes:** "Desi" humor, satire, and memes on platforms like TikTok and Instagram reflect cultural anxieties, global influences, and local trends.\n\n\n**Impact on Indian Society:**\n\n**Bridging Social Barriers:** Pop culture often transcends language barriers and brings people together, fostering a sense of shared experience. \n\n**Cultural Exchange & Reinterpretation:** Indian pop culture engages with international trends through adaptation and innovation, creating unique interpretations and cultural hybrids. \n\n\n**Evolving Consumerism:** The influence of pop culture drives consumer choices, from fashion and lifestyle to entertainment preferences.\n\n\n**Challenges:**\n\n**Representation & Inclusivity:** Pop culture in India still grapples with representation concerns around diverse backgrounds, gender roles, and socio-economic realities.\n\n**Cultural Sensitivity & Adaptation:** Balancing the appeal of international trends with the respect for local traditions and values remains an ongoing challenge. \n\n\n**Looking Forward:**\n\nIndia's pop culture landscape is dynamic and constantly evolving. The rise of digital platforms, increased internet penetration, and a growing sense of global citizenship are set to further expand its reach and influence. Understanding the complexities of this evolution will be crucial for navigating cultural nuances and appreciating the rich tapestry of Indian pop culture.\n\n\nLet me know if you'd like more details on any specific aspect mentioned above! \n', 'score': 1, 'iterations': 1}
(venv) prashantdp@Prashant:~/Desktop/Code/Projects/LangGraph/Agentic_AI$

```

Figure 3: Output of my silly little gemma2:2b

Solution 3

```

1 from langchain_mcp_adapters.client import MultiServerMCPClient
2
3 client = MultiServerMCPClient(
4     {
5         "gmail": {
6             "command": "npx",
7             "args": ["@gongrzhe/server-gmail-autoauth-mcp"],
8             "transport": "stdio",
9         }
10    }
11 )
12
13 from typing import TypedDict
14 from langgraph.graph import StateGraph, END
15 from langchain_groq import ChatGroq # fast
16 # from langchain_ollama import ChatOllama # slow af
17 from dotenv import load_dotenv

```

```

18 import asyncio
19 import os
20
21 load_dotenv()
22
23 # GROQ_API_KEY = os.getenv("GROQ_API_KEY")
24
25 llm = ChatGroq(model="llama-3.3-70b-versatile") #, api_key =
    GROQ_API_KEY)
26
27 class AgentState(TypedDict):
28     context: str # recipient name & email, sender name &
        email, context string
29     draft: str
30     score: int # heuristic for exiting the loop
31     iterations: int
32
33 def generate(state: AgentState):
34
35     prompt = f"""
36         Write a professional cold email based on the
37         context below.
38
39         Context:{state["context"]}
40
41         Sender Name: Prashant Meena
42
43         Requirements:
44         - Write only the final email body
45         - No placeholders (like [Name], YOUR_NAME,
46           COMPANY)
47         - No notes, explanations, or markdown
48         - Concise, personalized, and professional
49         - Sound natural and human, not robotic
50         - Clear reason for outreach
51         - End with a simple call-to-action
52         - Sign off with: Prashant Meena
53
54         This email will be sent exactly as generated, so
55         make it fully ready to send.
56     """
57
58     response = llm.invoke(prompt)
59
60     return {
61         "draft": response.content
62     }
63
64 def review(state: AgentState):
65     prompt = f"""
66     Review this email and score it from 1-10.

```

```

64
65     Email:
66     {state["draft"]}
67
68     Score based on:
69     - clarity
70     - professionalism
71     - grammar
72     - call to action
73
74     Return ONLY integer score.
75     """
76
77     response = llm.invoke(prompt)
78
79     try:
80         score = int(response.content.strip())
81     except:
82         score = 5
83
84     return {
85         "score": score,
86         "iterations": state["iterations"] + 1
87     }
88
89 def refine(state: AgentState):
90     prompt = f"""
91     Improve this email to get score 9+.
92
93     Current email:
94     {state["draft"]}
95     """
96
97     response = llm.invoke(prompt)
98
99     return {
100         "draft": response.content
101     }
102
103 def route(state: AgentState):
104     if state["score"] >= 8:
105         return "end"
106
107     if state["iterations"] >= 3:
108         return "end"
109
110     return "refine"
111
112 # initial -> generate -> review -> end($)
113 #         ^             |
114 #         |             |

```



```

115 # <-----
116
117 builder = StateGraph(AgentState)
118
119 builder.add_node("generate", generate)
120 builder.add_node("review", review)
121 builder.add_node("refine", refine)
122
123 builder.set_entry_point("generate")
124
125 builder.add_edge("generate", "review")
126 builder.add_edge("refine", "review")
127
128 builder.add_conditional_edges(
129     "review",
130     route,
131     {
132         "end": END,
133         "refine": "refine"
134     }
135 )
136
137 graph = builder.compile()
138
139 # sender = input(), we already know
140 recipient_name = input("Enter Recipient's Name: ")
141 recipient_email = input("Enter Recipient's Valid Gmail: ")
142 email_context = input("Provide some context regarding Email: ")
143
144
145 initial_state = {
146     "context": f"""
147         recipient_name: {recipient_name},
148         recipient_email: {recipient_email},
149         email_context:{email_context}
150     """,
151     "draft": "",
152     "score": 0,
153     "iterations": 0
154 }
155 result = graph.invoke(initial_state)
156
157 async def send(final_draft: str): # USES MCP SERVER
158     global recipient_name, recipient_email
159
160     tools = await client.get_tools()
161
162     print("Available tools:")
163     for tool in tools:
164         print(tool.name)
165

```

```

166         for tool in tools:
167             if "send" in tool.name.lower():
168                 response = await tool.ainvoke({
169                     "to": [recipient_email],
170                     "subject": "Quick Introduction",
171                     "body": final_draft
172                 })
173                 print(response)
174                 return
175
176         print("No send tool found.")
177
178 asyncio.run(send(result["draft"]))

```

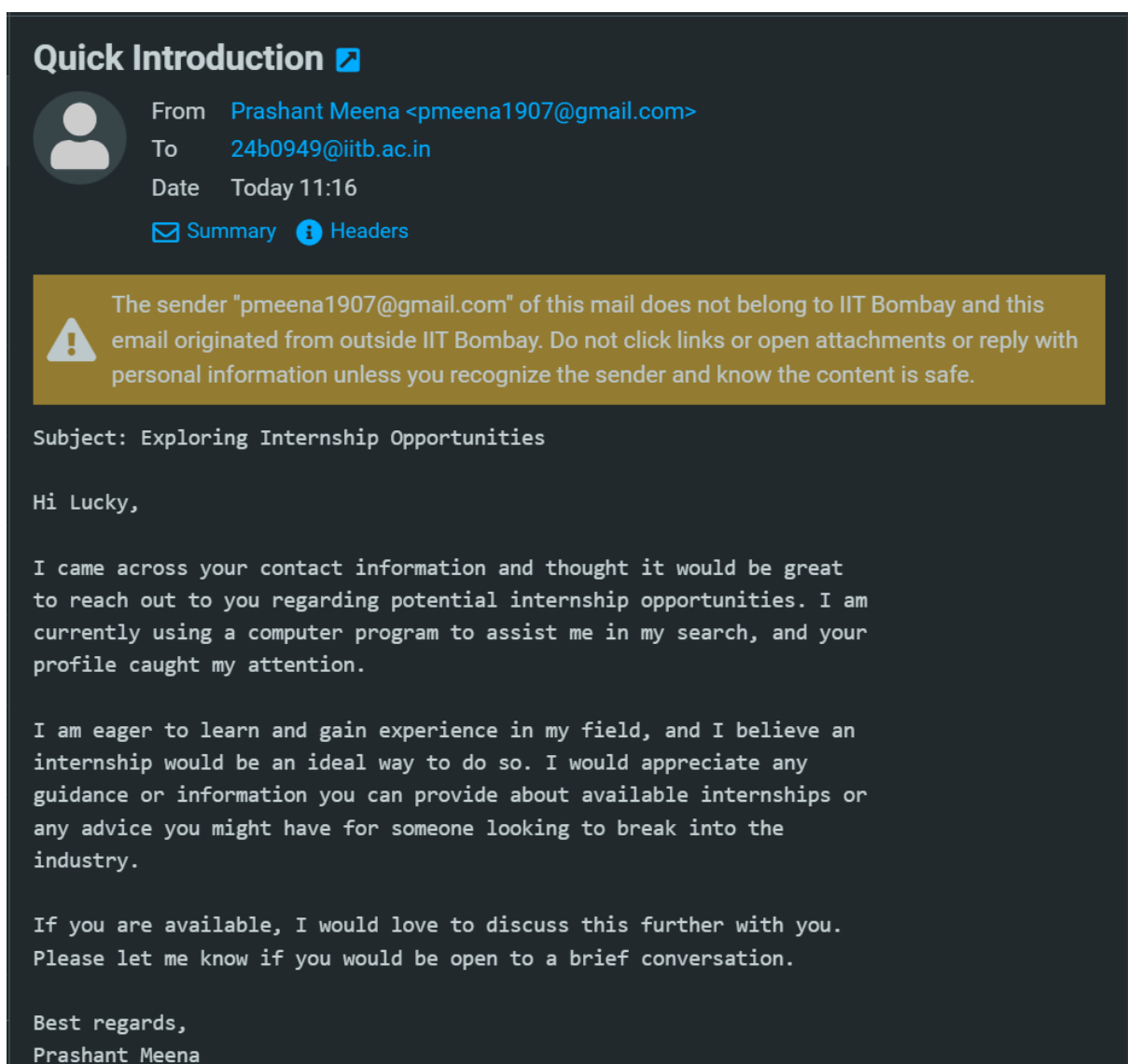


Figure 4: EMail Proof